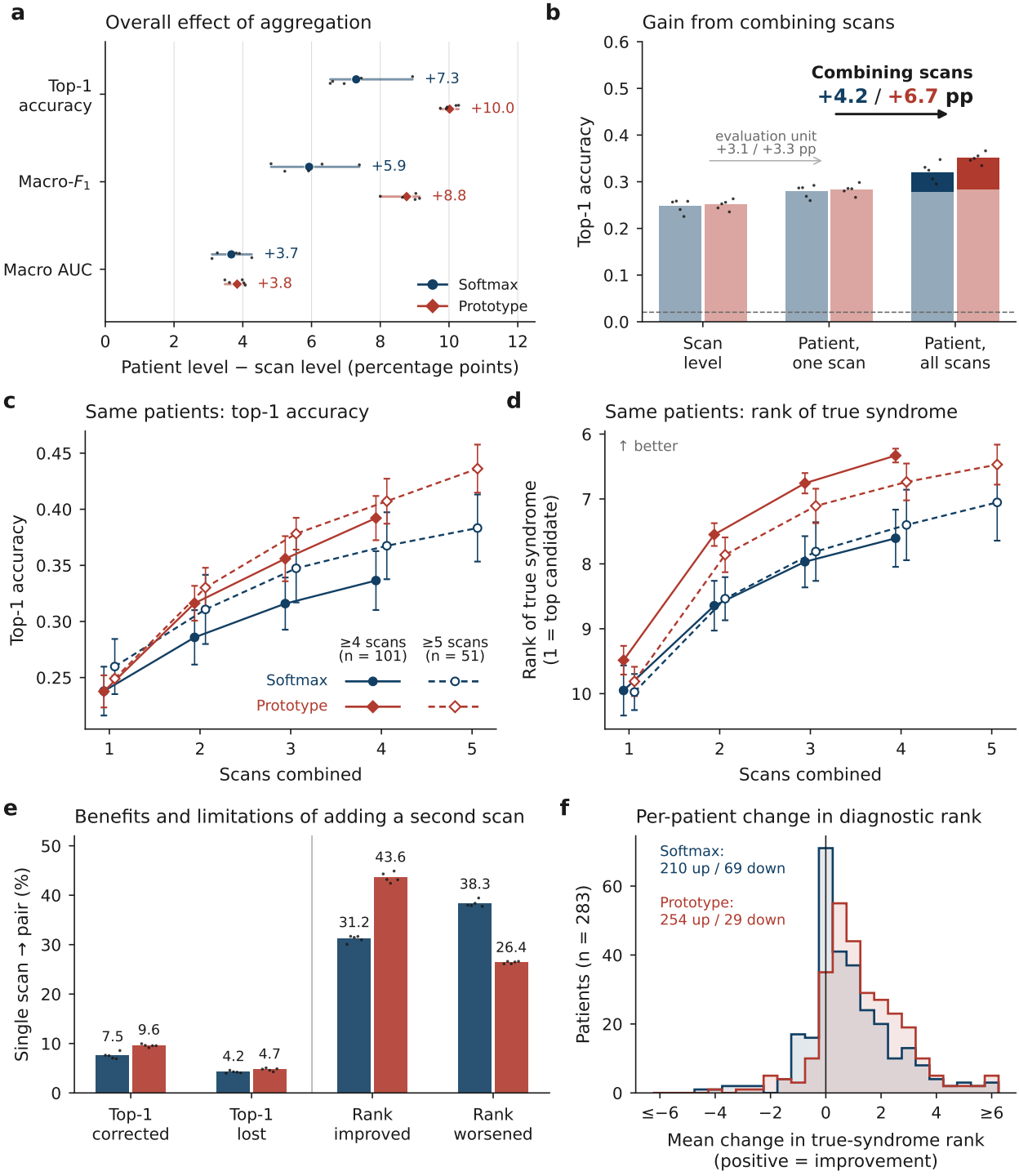
a
Overall effect of aggregation
Top-1 accuracy
Macro-F_1
Macro AUC
+7.3
+10.0
+5.9
+8.8
+3.7
+3.8
Softmax
Prototype
Patient level − scan level (percentage points)
b
Gain from combining scans
Combining scans
+4.2 / +6.7 pp
evaluation unit
+3.1 / +3.3 pp
Top-1 accuracy
Scan level
Patient, one scan
Patient, all scans
c
Same patients: top-1 accuracy
Top-1 accuracy
≥4 scans (n = 101)
≥5 scans (n = 51)
Softmax
Prototype
Scans combined
d
Same patients: rank of true syndrome
↑ better
Rank of true syndrome (1 = top candidate)
Scans combined
e
Benefits and limitations of adding a second scan
Single scan → pair (%)
7.5
9.6
4.2
4.7
31.2
43.6
38.3
26.4
Top-1 corrected
Top-1 lost
Rank improved
Rank worsened
f
Per-patient change in diagnostic rank
Softmax:
210 up / 69 down
Prototype:
254 up / 29 down
Patients (n = 283)
Mean change in true-syndrome rank
(positive = improvement)